

Chapter 5

Relationships in Triangles

Unit 2 — Triangles

McGraw Hill Glencoe Geometry Texas, 2015

Key Vocabulary

Term	Definition
Perpendicular bisector	$\perp$ to a segment at its midpoint
Angle bisector	Ray dividing an angle into two congruent angles
Median	Segment from vertex to midpoint of opposite side
Altitude	Perpendicular segment from vertex to opposite side (or its extension)
Circumcenter	Intersection of perpendicular bisectors
Incenter	Intersection of angle bisectors
Centroid	Intersection of medians
Orthocenter	Intersection of altitudes

Triangle Centers

Center	From	Property
Circumcenter	Perp. bisectors	Equidistant from vertices; circumscribed circle center
Incenter	Angle bisectors	Equidistant from sides; inscribed circle center
Centroid	Medians	Balancing point; $\frac{2}{3}$ of median from vertex
Orthocenter	Altitudes	May be inside, on, or outside triangle

Theorem 5-1: Perpendicular Bisector Theorem

If a point is on the **perpendicular bisector** of a segment, it is **equidistant** from the endpoints.

$$P \text{ on perp. bisector of } \overline{AB} \Rightarrow PA = PB$$

Theorem 5-2: Converse of Perpendicular Bisector

If a point is **equidistant** from the endpoints of a segment, it lies on the **perpendicular bisector**.

$$PA = PB \Rightarrow P \text{ lies on perp. bisector of } \overline{AB}$$

Theorem 5-3: Angle Bisector Theorem

If a point is on the **bisector of an angle**, it is **equidistant** from the sides of the angle.

$$P \text{ on bisector of } \angle A \Rightarrow d(P, \overrightarrow{AB}) = d(P, \overrightarrow{AC})$$

Theorem 5-4: Converse of Angle Bisector Theorem

If a point in the interior of an angle is **equidistant from the sides**, it lies on the **angle bisector**.

Theorem 5-5: Circumcenter Theorem

The **circumcenter** of a triangle is equidistant from all three vertices.

$$CA = CB = CC$$

The circumcenter is the center of the circumscribed circle.

Theorem 5-6: Incenter Theorem

The **incenter** of a triangle is equidistant from all three sides.

The incenter is the center of the inscribed circle.

Theorem 5-7: Centroid Theorem

The **centroid** divides each median in a 2 : 1 ratio from vertex to midpoint.

$$\text{Centroid to vertex} = \frac{2}{3} \times \text{median length}$$

Example: If median $AM = 12$, then centroid G satisfies $AG = 8$ and $GM = 4$.

Theorem 5-8: Exterior Angle Inequality

An **exterior angle** of a triangle is greater than either of the **non-adjacent interior angles**.

$$m\angle 4 > m\angle 1 \quad \text{and} \quad m\angle 4 > m\angle 2$$

Theorem 5-9: Angle-Side Relationship

In a triangle, the **larger side** is opposite the **larger angle**.

$$m\angle A > m\angle B \Rightarrow BC > AC$$

(The longest side is opposite the largest angle.)

Theorem 5-10: Converse of Angle-Side Relationship

In a triangle, the **larger angle** is opposite the **larger side**.

$$BC > AC \Rightarrow m\angle A > m\angle B$$

Theorem 5-11: Triangle Inequality Theorem

The **sum of any two sides** of a triangle must be **greater than** the third side.

$$AB + BC > AC$$

$$AB + AC > BC$$

$$BC + AC > AB$$

Test: Can 5, 7, 10 form a triangle?

$5 + 7 = 12 > 10 \checkmark$, $5 + 10 = 15 > 7 \checkmark$, $7 + 10 = 17 > 5 \checkmark \rightarrow$ **Yes**

Example — Triangle Inequality

Can segments of length 3, 5, and 9 form a triangle?

$$3 + 5 = 8 \not> 9 \quad \times \text{ FAILS}$$

No, these cannot form a triangle.

Theorem 5-12: Hinge Theorem (SAS Inequality)

If two triangles have two pairs of congruent sides, the triangle with the **larger included angle** has the **longer third side**.

$$\overline{AB} \cong \overline{DE}, \overline{BC} \cong \overline{EF}, m\angle B > m\angle E \Rightarrow AC > DF$$

Theorem 5-13: Converse of Hinge Theorem

If two triangles have two pairs of congruent sides, the triangle with the **longer third side** has the **larger included angle**.

$$AC > DF \Rightarrow m\angle B > m\angle E$$

Example — Centroid

Triangle ABC has centroid G . Median $\overline{BD}$ has length 18.

$$BG = \frac{2}{3}(18) = 12 \qquad GD = \frac{1}{3}(18) = 6$$

Chapter 5 – Theorem Summary

#	Name	Key Idea
Thm 5-1/2	Perp. Bisector	On bisector $\iff$ equidistant from endpoints
Thm 5-3/4	Angle Bisector	On bisector $\iff$ equidistant from sides
Thm 5-5	Circumcenter	Equidistant from vertices
Thm 5-6	Incenter	Equidistant from sides
Thm 5-7	Centroid	$\frac{2}{3}$ from vertex along median
Thm 5-8	Ext. Angle Ineq.	Exterior $\angle >$ non-adjacent interior $\angle$ s
Thm 5-9/10	Angle-Side	Larger $\angle \iff$ longer opposite side
Thm 5-11	Triangle Inequality	Sum of any 2 sides $>$ 3rd side
Thm 5-12/13	Hinge Theorem	Larger included $\angle \iff$ longer 3rd side